End-to-End Performance Modelling and Analysis

Communication Networks • Continuous Assessment 1
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Declared assumptions • Reproducible simulations • Measured uncertainty

Three tasks, one image journey

01

Graph theory
Topology and routing

02

Wireless access
Rate and polling delay

03

Queueing theory
M/M/1 and delay limits

End-to-end time = polling wait + grant + transmission + backhaul + MEC waiting and processing.

The numerical inputs are explicit assumptions

4 cameras

Independent Poisson arrivals

1 Mbit

Fixed-size baseline images

20 MHz

Bandwidth; SNR = 10 dB

50 images/s

Mean MEC service rate

100 ms

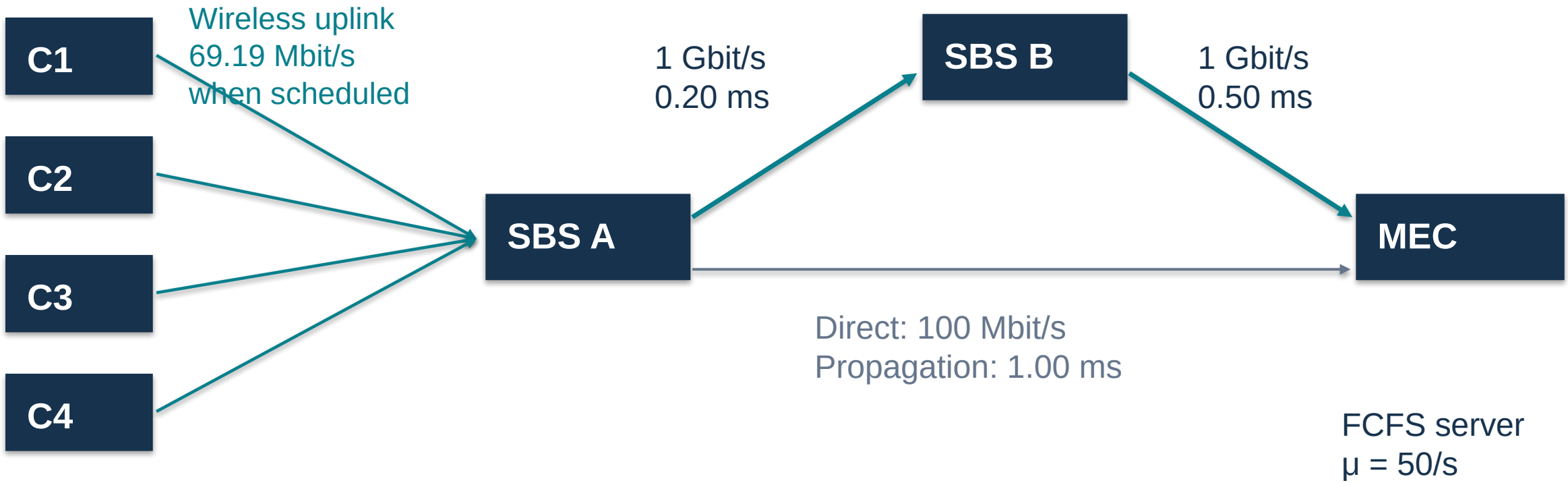
Chosen mean-time target

0.20 ms

Overhead per radio visit

λ = total images/s across four cameras; each camera generates λ/4.

A graph describes the available paths



Camera nodes: λ/4 and image size.
Edge weights: payload serialization + propagation.

A faster route can have an extra hop

11.00 ms

Direct wired backhaul: A → MEC

2.70 ms

Selected wired backhaul: A → B → MEC

Selected route: C1 → A → B → MEC
Payload + propagation: 17.15 ms; saving: 8.30 ms

Route weight excludes the grant, polling wait and MEC processing.

Shannon gives an ideal scheduled rate

$$R = W \log_2(1 + \gamma)$$

SNR = 10 dB → γ = 10; W = 20 MHz

69.19 Mbit/s

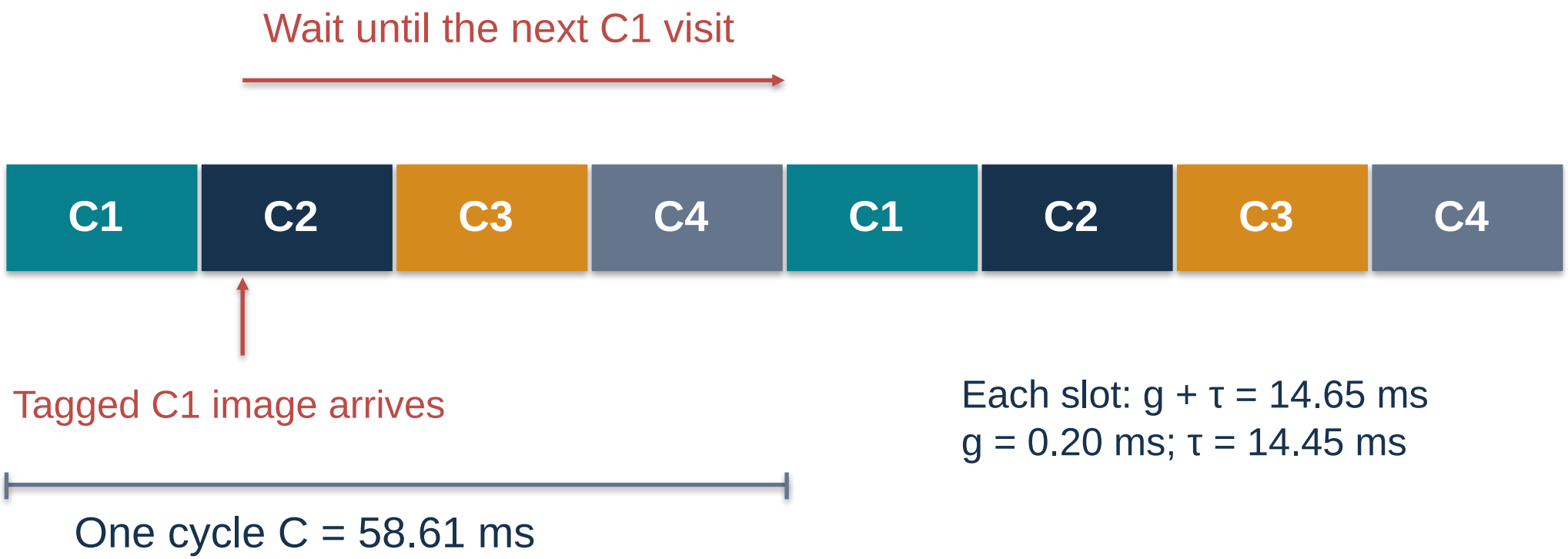
Ideal rate while scheduled

14.45 ms

Payload duration τ = S/R

OFDMA RBs are granted to one camera at a time in this model.

Random arrival phase creates polling wait



Each slot: $g + \tau = 14.65$ ms
$g = 0.20$ ms; $\tau = 14.45$ ms

Mean tagged wireless time = $C/2 + g + \tau = 43.96$ ms

Backlog increases the camera waiting time

Mean polling wait = C / [2(1 – r)]

$r = (\lambda/N)C = \lambda(g + \tau)$
Radio stability: $r < 1$, or $\lambda < 68.24$ images/s

46.25 ms

Polling wait at total $\lambda = 25$/s

70.81 ms

Polling wait at total $\lambda = 40$/s

Image-size variability changes transmission service

X = image size in bits; S = E[X]; V = X/R is transmission time.

Fixed image size

X = S
V = S/R is constant
Transmission queue: M/D/1

Exponential image size

X ~ Exp(rate 1/S)
V ~ Exp(rate R/S)
Transmission queue: M/M/1

Queue labels assume Poisson input and a continuously available server.
Fixed overhead and cyclic polling modify that simple model.

Simulation tests each stage and the whole chain

30

Independent runs per arrival rate

100,000

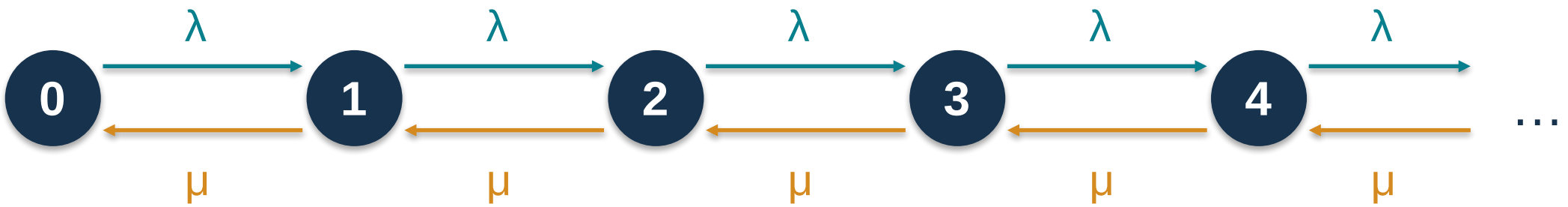
Measured images in each run

20,000

Warm-up images discarded

11 arrival rates • Fixed seed • Pointwise 95% intervals
10 deterministic tests passed; replication summaries included.

Little’s law gives the M/M/1 delay bound



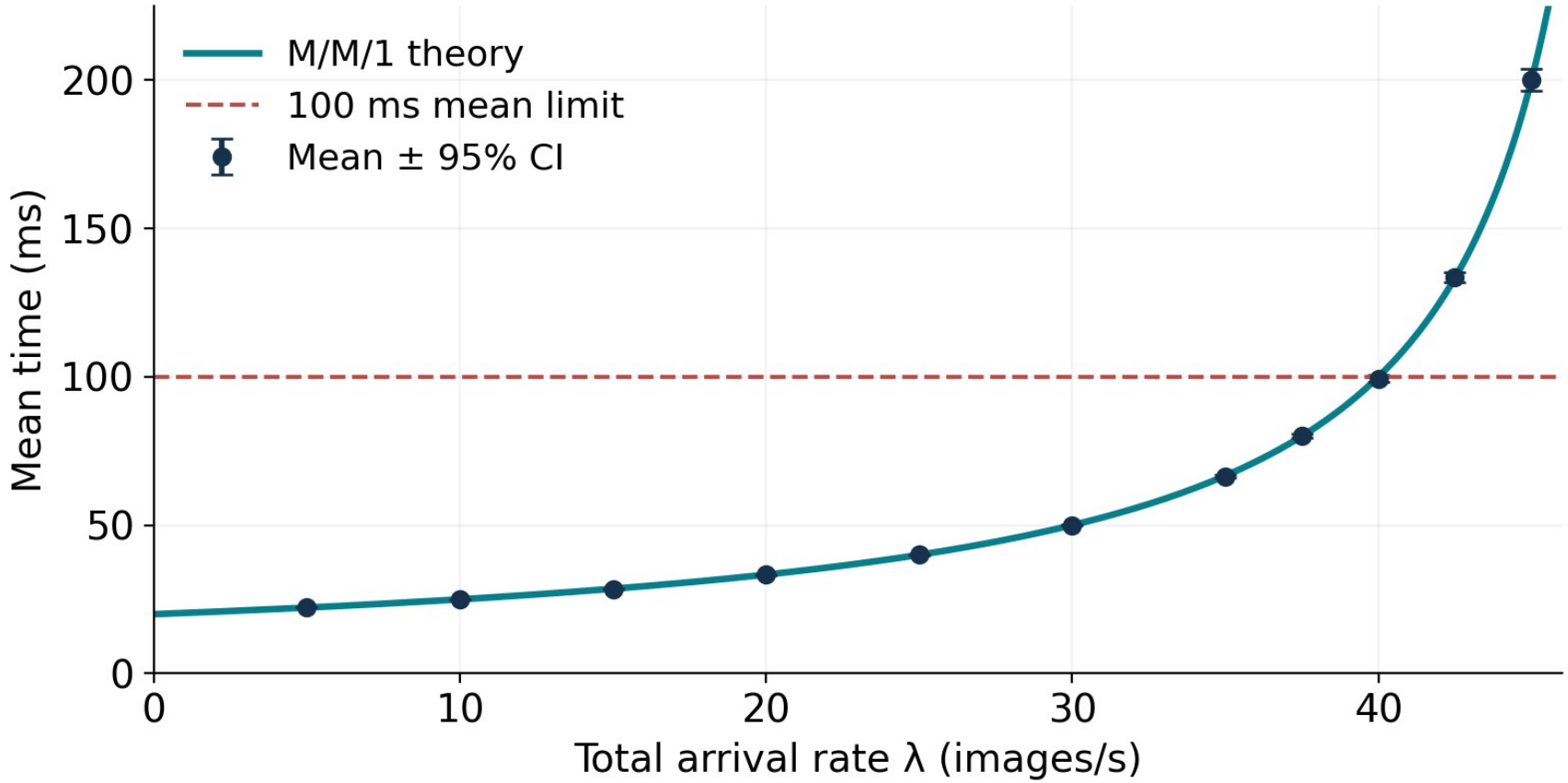
$L_s = \rho/(1-\rho) = \lambda T \mid \rho = \lambda/\mu < 1$

T = waiting + processing = $1/(\mu - \lambda)$

$\lambda_{max} = \mu - 1/T_{max}$ = 40 images/s

μ = 50/s; T_{max} = 0.100 s. A positive bound requires $T_{max} > 1/\mu$.

The isolated server follows the theoretical curve



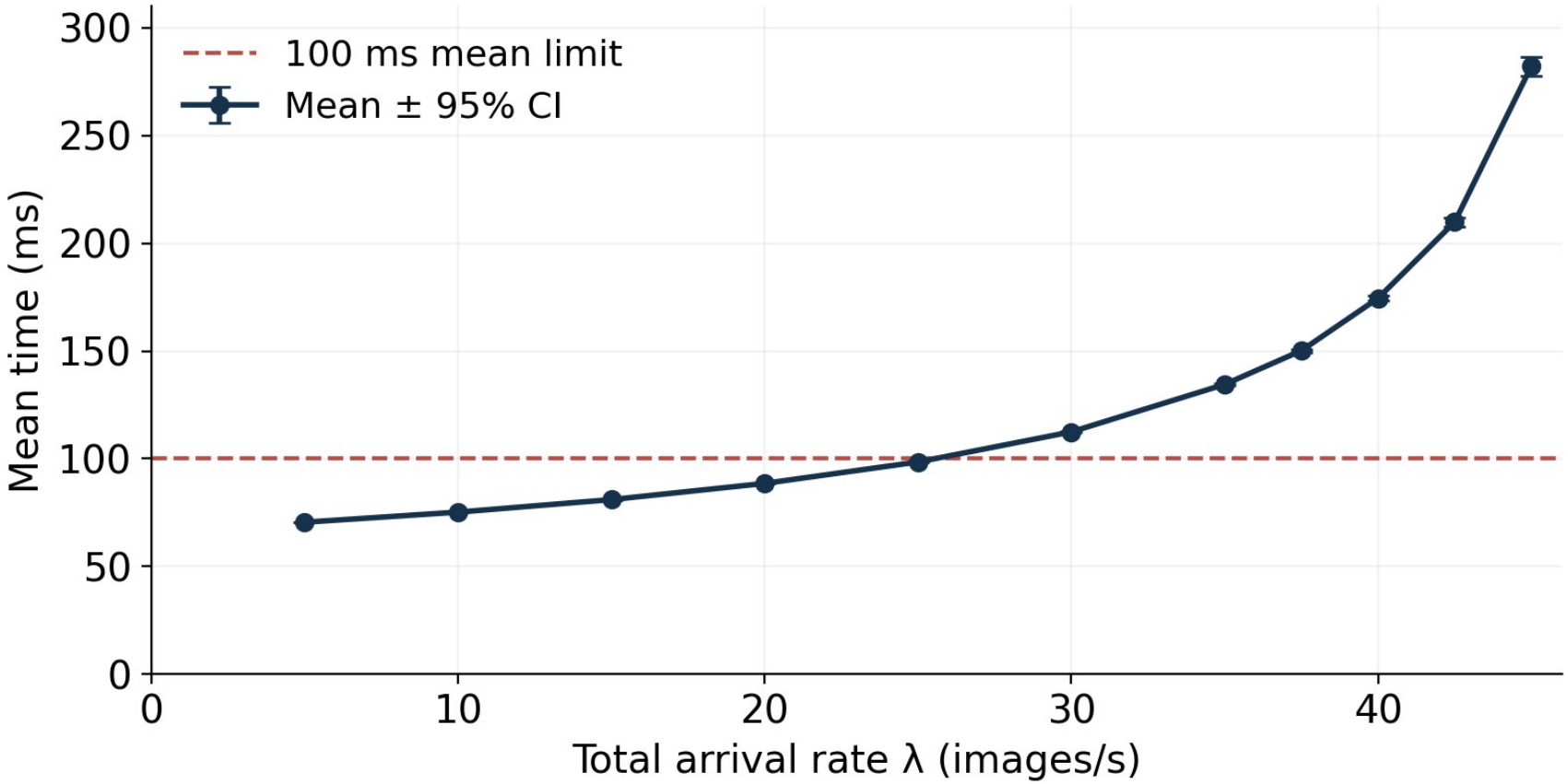
0.62%

Largest relative mean-time error

At λ = 40/s:
Theory: 100 ms
Simulation: 99.38 ms

10/11 pointwise CIs cover theory.

End-to-end delay includes the upstream stages



25 images/s

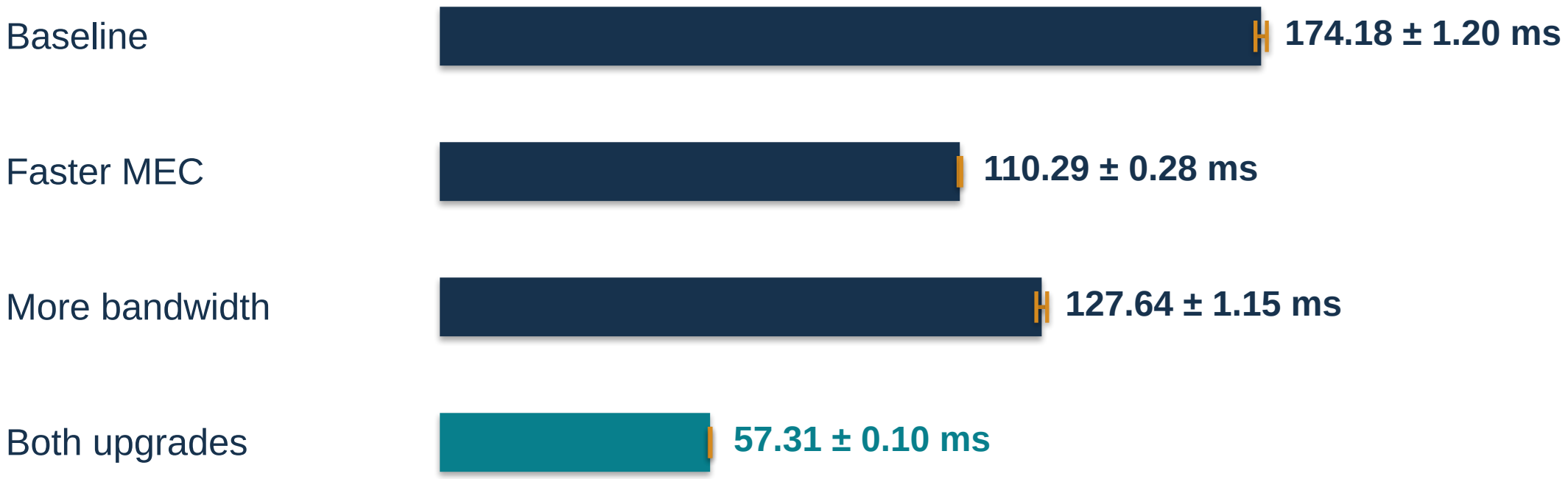
98.52 ms

30 images/s

112.48 ms

25/s is the highest tested passing rate, not an exact limit.

Radio and processing should be upgraded together



Mean ± 95% CI; SNR stays at 10 dB. Combined upgrades: 9.51% exceed 100 ms.

Design for the whole image journey

01 Select routes using capacity and delay.

02 Include camera polling and MEC queueing.

03 Check mean delay and deadline reliability separately.

Questions

Sources: assessment brief; course Chapter 2; Shannon (1948); Dijkstra (1959); Little (1961). Full references and derivations are in the report.